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RESEARCH ARTICLE

Understanding the Molecular Level Interaction of Lithium-ions With Cellulose Based Eutectogel: A Molecular Dynamics Study

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ABSTRACT

The development of biocompatible polymer-based solid-state electrolytes represents a promising direction for advanced energy storage applications. This work provides the first atomistic comparison of Li-ion transport in Succinonitrile (SN)-based eutectogels containing Lithium Difluoro(oxalate)borate (LiDFOB) and Lithium bis(trifluromethanesulfonyl)imide (LiTFSI) dual-salts across three distinct derivatives: Cyanoethyl (CEC), Ethyl (EC), and Methyl (MC) cellulose. The structural analysis reveals that the CEC system demonstrates enhanced molecular ordering through increased hydrogen bonding interactions, which suggests the higher mechanical stability of the electrolyte. The MC systems exhibit stronger lithium-ion interactions with both the polymer matrix and electrolyte environment, which is due to the less stearic hindrance. The diffusion analysis indicates that lithium ions associated with LiTFSI in the EC system show diffusive behavior with a self-diffusion coefficient of 1.44×10^{-12} $m^2.s^{-1}$. These molecular-level insights into structural organization and ion transport mechanisms provide valuable design criteria for developing high-performance, biocompatible solid-state electrolytes for next-generation lithium-ion batteries.

1 | Introduction

Eutectogels are the state-of-the-art inclusion of polymeric gel materials that have multidisciplinary applications because of their unique features [1–5]. Euetctogel is primarily composed of a polymeric framework and a deep eutectic solvent (DES) [1]. The Hydrogen Bond Donors (HBDs) and Hydrogen Bond Acceptors (HBAs) are the constituents of DES, which show the unique qualities like low vapor pressure, highly sustainable, and majorly high conductivity values [1, 6–9]. The fascinating properties of DESs are also exported to eutectogels, resulting in significant enhancement compared to contemporary gels. For example, natural deep eutectic solvent (NADES) is seen to be a more improved option for drug delivery vehicles than hydrogels [10]. The wastewater treatments are efficiently carried out by the eutectogels made up of DES and L-Amino acids [7]. The eutectogels are modified for anti-drying and anti-swelling applications [11–12]. The eutectogels prepared with poly(vinyl alcohol) (PVA) polymer, which is a sustainable and greener option, show high strength along with anti-freezing properties [1, 5]. Recently, to address the safety difficulties involving Lithium-Ion batteries (LIBs) and to boost their conducting capacity, the eutectogel is considered as a potential candidate [4–5, 13]. Herein, contemporary gels like organogel and hydrogel face the issues of low conductivity, toxicity, and overall, very low performance [14, 15]. The entrapment of ionic liquid in a polymeric matrix creates an ionogel that shows high ionic conductivity, thermal stability, high mechanical strength, and sustainability [5, 16]. However,

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the expensive precursors and complex purification methodology make ionogels unattractive for commercial application and thus put the searchlight on eutectogel, which so far mitigates this criterion [17, 18].

Various classes of DESs include choline class [19], quaternary ammonium salt-based [20, 21], metal salt-based [22, 23], carboxylic acid-based [23], sugar-based [24], and alcohol-based [9, 25]. Among them, Lithium bis(trifluromethanesulfonyl)imide (LiTFSI) based DES is the notable choice for the eutectogel preparation. The availability of lithium (Li^+) ions in the electrolyte enhances the Li^+ ion transportation in the potential LIB application [26, 27]. However, to enhance the strength of the eutectogel, polymeric matrices, mostly biocompatible materials such as lignin, CEC, α-cellulose, hydroxyethyl cellulose, and carboxymethyl cellulose, are often infused [26, 28–30]. The inclusion of biocompatible polymers widens the scope of application of eutectogel other than LIBs [16, 31–33]. The eutectogel prepared by adding LiTFSI ions, N-methylphthalimide (NMePh) mixed with urea, shows improved electrochemical reversibility as well as stability. NMePh contributes for the improvement of reversibility, and urea participates in stabilising the NMePh anions and to reduce the viscosity of the system to make ionic motion less hindered [34]. The use of biocompatible substances reduces the toxicity associated with eutectogels. The eutectogel prepared by cotton cellulose infusion to the mixture of acrylic acid and zinc chloride ($ZnCl_2$) shows enhancement in ionic conductivity as well as mechanical strength equivalent of ionogels and hydrogels [27]. DESs, prepared from metal salts like lithium chlorides (LiCl) and $ZnCl_2$ with ethylene glycol as a HBD, provide stability and eco-friendliness. PVA interacts with DESs via hydrogen bonds (H-bonds) to form eutectogels with enhanced properties. The addition of lignin significantly improves mechanical strength, while glycerol regulates PVA crystallinity. These gels show excellent conductivity (19.7 mS cm^{-1}), elongation (550%), and stability at −20°C [5]. Due to the presence of many functional moieties, the formation of H-bonds, Van der Waals interactions, and long-range weak interactions has a major effect on ion transportation through the electrolyte. However, the experimentation is often unable to explore such atomistic details, leading to the knowledge gap on ion distribution and transportation in eutectogel.

The study of interactions at the atomistic level and precise calculations is made possible by molecular dynamics (MD) simulations, which help to uncover the difference between the atomic electrolyte properties and bulk properties [35, 36]. MD simulations are used widely to study the long- and short-range interactions among molecules [37–39]. The structural properties, as well as transport properties like viscosity and diffusivity of the systems, can be determined and studied through the analysis of the trajectory [35, 40, 41]. The interaction and binding energies of the eutectogel are helpful to find the compatible eutectogel for the electrolyte in the LIB application. The eutectogel is designed on the basis of poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP) organic-polymer gels, showing less resistance to mechanical stress and high polarization quality, which makes it a fascinating electrolyte for lithium-ion batteries [26]. It is observed through MD simulation that Li^+ ions derived from the LiTFSI salts form clusters and their size increases with the increase in the concentration of LiTFSI salt for poly(diethyleneoxide-alt-oxymethylene), denoted as P(2EO-MO) polymer [26]. Using Density Functional Theory (DFT) and MD simulations, DESs are integrated with inorganic silica gel (SiO_2-gel) and organic PVDF-HFP matrices. DES, composed of LiTFSI as the HBA and N-methylacetamide (NMAc) as the HBD in a 1:4 ratio, forms ETG1 (SiO_2-gel matrix) and ETG2 (PVDF-HFP matrix). Structural integrity is evaluated through binding energy, interaction energy, H-bonding, and HOMO-LUMO analyses, revealing higher stability, lower polarization, and greater mechanical resistance for ETG1 compared to ETG2 [42]. MD simulations highlight the importance of H-bonding, radial distribution functions (RDF), and mean square displacement (MSD) in determining ionic conductivity and lithium-ion mobility [35]. Lignin addition to a PVA-based eutectogel, formed with DES containing LiCl and ethylene glycol (1:5), enhances structural strength via increased H-bonding, improves lithium-ion diffusivity, and influences interactions with chloride ions, as confirmed by RDF and MSD analyses [36]. Molecular docking and quantum chemical calculation is used to study the interactions between the lithium-ion to the hydroxypropyl cellulose (HPC), hydroxypropylmethyl cellulose (HPMC), and hydroxyethyl cellulose (HEC). The result shows that HPMC-based electrolyte shows higher ionic conductivity due to its unique molecular structure [43]. The MD simulation is used to study the HEC-based eutectogel. Due to the strong H-bonding with the electrolyte, it shows higher ionic conductivity [26]. MD simulations are used to study methyl and carboxymethyl cellulose-based eutectogels, revealing the molecular hopping mechanism of lithium ions [44]. Hence, MD studies on the limited number of eutectogel systems elucidate the atomistic dynamics and atomistic level interactions that leads the mobility of lithium-ions, which is immensely helpful in the design specification of LIBs.

In this study, organic polymer-based eutectogels, which are made up of Succinonitrile (SN), LiTFSI, and Lithium Difluoro(oxalate)borate (LiDFOB) are considered for MD simulation. For the sake of mechanical strength enhancement, Cyanoethyl cellulose (CEC), Ethyl Cellulose (EC), and Methyl Cellulose (MC) are taken into account as organic polymers. The effect of different polymers on the Li^+ ions transportation through the system is the primary objective of this work. The comprehensive assessment of structural orientation of polymers and ionic constituents has been determined with the help of H-bonding, RDF, and *CNs*. The mechanism of ion transportation has been explored with the analysis of MSD curves and ionic conductivity. The analysis of H-bonding indicates nonbonded association of various constituents of eutectogel. RDF and *CN* become very effective tool to identify the interacting moieties. Together with the diffusivity, the ionic conductivity values are a direct indicator of feasible ion transportation through the electrolyte. The atomistic interaction and transportation details observed from this study will elucidate the design criterion of eutectogels for LIBs.

2 | Computational Methodology

2.1 | Details of Systems

For this study, DES comprising of SN, LiTFSI is updated by the addition of LiDFOB. The organic matrices, like CEC, EC, and MC, are infused with the updated DES to prepare the eutectogels. Thus, three eutectogel systems consist of 3 different organic

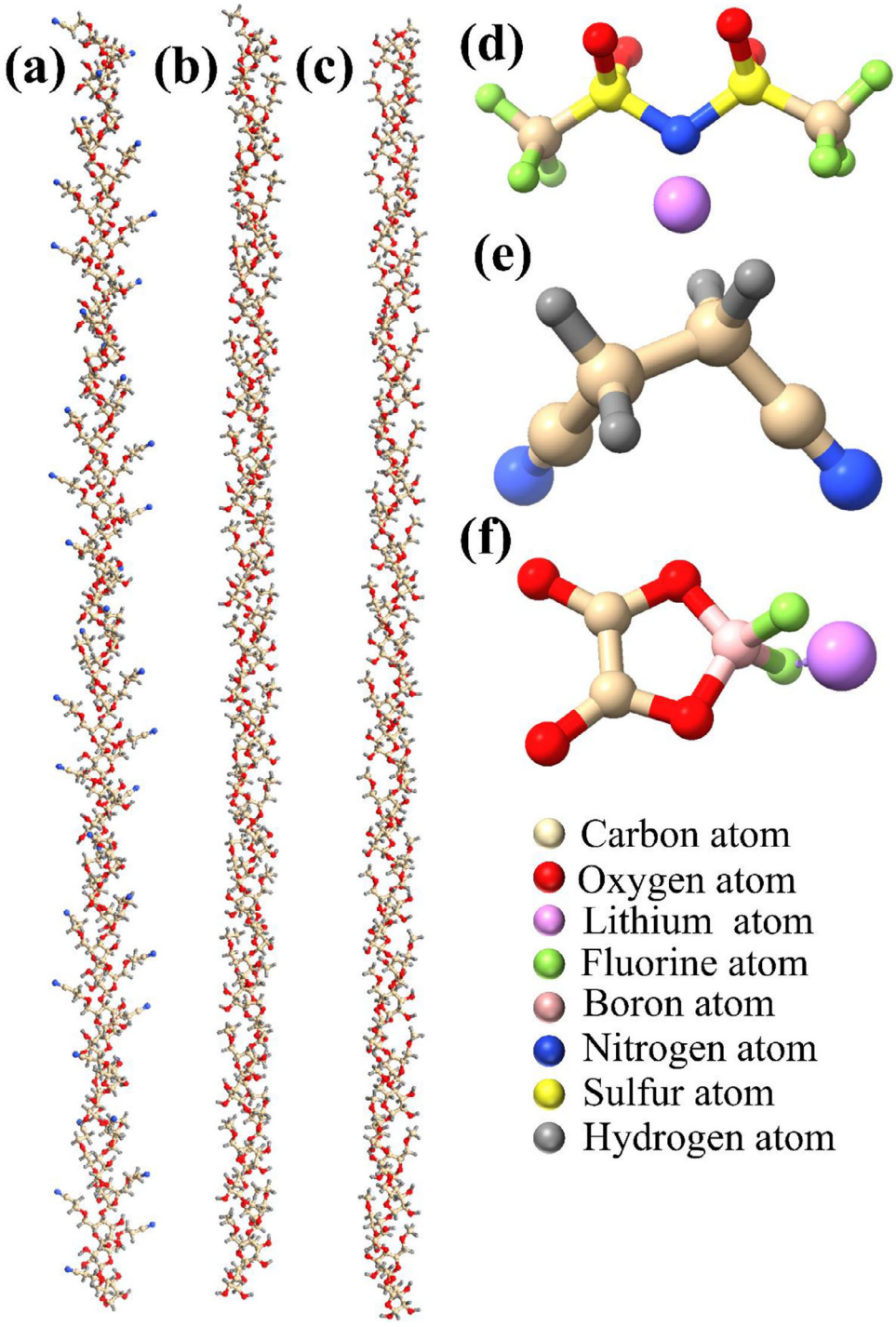


FIGURE 1 | Initial structure of molecules present in electrolyte (a) CEC, (b) EC, (c) MC, (d) LITFSI, (e) SN, (f) LIDFOB.

matrices that are formed. The configurational details of the systems are included in Table S1 of the supplementary material. The Avogadro open-source package is used to draw the initial structures of all the components of eutectogels [45]. Figure 1 depicts the configurational representations of the components. Using the AM1-BCC technique, the partial charges of each moiety are calculated [46, 47].

The Optimized Potentials for Liquid Simulations (OPLS) force field is used for the accurate analysis of nonbonded interactions (i.e., van der Waals and coulombic) and bonded interactions among the components of the system [48, 49].

$$V_{total} = \sum_{bonds} k_r\left(r - r_{eq}\right)^2 + \sum_{angles} k_\theta\left(\theta - \theta_{eq}\right)^2 + \sum_{dihedrals} \times k_\chi\left[1 + \cos\left(n\chi - \delta_\chi\right)\right] + \sum_{improper} k_\psi\left[1 + \cos\left(n\psi - \delta_\psi\right)\right] + \sum_{i=1}^{N-1}\sum_{i<j}^{N}\left\{4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] + \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}}\right\} f_{ij} \quad (1)$$

Here, Equation (1) is the functional representation of OPLS force field where, V_{total} is the total potential energy contained by the system. It includes terms like stretching energy exhibited due to bonds, bending energy exhibited by angles, tortional energy exhibited by improper as well as proper dihedrals. Energy derived from nonbonded interactions involving van der Waals interactions and Coulombic interactions. k_r, k_θ, k_χ and k_ψ represent constants considered in bond stretching calculation term, bending of angles calculation term, proper dihedral angle calculation between two planes and improper dihedral angle calculation respectively. r and θ show length of bond and angles between the molecules respectively. r_{eq} and θ_{eq} denote standard bond length and standard angle between the molecules respectively. ε_{ij} and σ_{ij} are the L-J potential parameters between i^{th} and j^{th} atoms. q_i is the charge of ith atoms. q_j is the charge of jth atoms. δ_χ and δ_ψ denote phase shift terms for proper dihedrals and improper dihedrals respectively. ε_0 represent permittivity value of vacuum. r_{ij} denotes the distance between i^{th} and j^{th} atoms. The term n represents multiplicity. χ and ψ denotes the proper dihedral angle and improper dihedral angle between the planes, respectively. The f_{ij} values are provided below.

$$f_{ij} = \begin{Bmatrix} 0.5, if, i, j = 1, 4 \\ 1.0, otherwise \end{Bmatrix} \tag{2}$$

2.2 | MD Simulation Framework

GROMACS [50] has been used to perform MD simulations. Initially, an energy minimization step is carried out to achieve a stable composition by reducing potential energy of system. The number of steps performed 5×10^6. The step size considered is 10^{-3} nm. 1000 kJ/mol nm is the convergence threshold value. The energy minimization stage is carried out using the Steepest Descent algorithm. After the energy minimization step, the equilibration step is performed using the linear constant solver (LINCS) algorithm, which helps to control the vibration of H-bonds in the system [51]. During this step, Leap-frog integrator is used to integrate the motion equation at a time step of 2 fs. To manage the van der Waals interactions, Particle-Mesh Ewald (PME) has been utilized, maintaining cutoff distances of 1.0 and 1.2 nm. [52]. Further, this equilibration divides into 2 ensembles named as NVT and NPT ensemble. An annealing process is added to the NVT step to create a uniform environment for better performance. The NVT annealing process includes two steps, where the first step is temperature rising from 300 to 500 K, and the second step is cooling the system by changing temperature from 500 to 300 K. The temperature rising is carried out at a rate of 0.4 and 0.1 K/ps is the rate for the cooling step. NVT ensemble performed for 30 ns time duration, and NPT ensemble for 10 ns to achieve an anticipated density of the system. Finally, the production run is executed for 30 ns. V-rescale thermostat [53] is used to uphold the temperature throughout equilibration and production runs, as well as for the pressure Parrinello-Rahman barostat [54]. Visual Molecular Dynamics (VMD) [55] is employed for the scrutiny of MD-generated trajectory curves. The time-dependent simulation snapshots are given in Figure 2 and Figures S1–S3 of supplementary materials (SM).

2.3 | Analysis

2.3.1 | Radial Distribution Function and Coordination Number

The distribution of different ions, atoms and molecules in the system is studied using RDF as well as the *CN*. The expressions for RDF and *CN* are expressed in Equations (3) and (4).

$$g_{PQ}(r) = \frac{\langle \rho_Q(r) \rangle}{\langle \rho_Q \rangle_{local}} = \frac{1}{\langle \rho_Q \rangle_{local}} \frac{1}{N_P} \sum_{i \in P}^{N_P} \sum_{j \in Q}^{N_Q} \frac{\delta(r_{ij} - r)}{4\pi r^2} \tag{3}$$

$$CN = 4\pi\rho N \int_0^r g(r) r^2 dr \tag{4}$$

where, Equation (3) includes term $\delta(r_{ij} - r)$ known as Dirac Delta function in which r_{ij} is the distance between i^{th} and j^{th} atoms and r is the distance between other atom and central atom. N_P represents the total numerical value of atoms belongs to P.N_Q represents the total numerical value of atoms belongs to Q. The local average number density of atoms is given by $\langle \rho_Q \rangle_{local}$ and $\langle \rho_Q(r) \rangle$ denotes the same at a distance r. Equation (4) is obtained by integrating RDF function $g_{PQ}(r)$. N, ρ and r denote number of atoms present in the system, number of density of atoms and radius of shell when reference atom is assumed at center respectively.

2.3.2 | Hydrogen Bond

The H-bond analysis helps to understand the structural integrity of the system over a period of time. The following expression, Equation (5) is used to find the number of H-bonds represented by C_{HB}.$H(\tau + t)$ and $H(\tau)$ are the H-bond order parameters at time equal to $(\tau + t)$ and τ respectively.

$$C_{HB} = \frac{\langle H(\tau) H(\tau + t) \rangle}{\langle H(\tau) H(\tau) \rangle} \tag{5}$$

2.3.3 | Mean Square Displacement

The study of diffusivity has the major importance with respect to conductivity analysis. Here, the MSD plots give an idea about the diffusion behavior of ions in the system, which is denoted by β. The β value equal to unity suggest the diffusive diffusion behavior of ions, β value less than unity and β value greater than unity are the indications of sub-diffusive and super-diffusive behavior of the ions, respectively. The expression for the β is deduced from Einstein relation [59] reported as Equation (6).

$$\lim_{t \to \infty} \left\langle \| r_i(t) - r_i(0) \|^2 \right\rangle_{i \in P} = 6 D_A t \tag{6}$$

$$\beta = \lim_{t \to \infty} \frac{d\left[\log\left(\Delta r^2\right)\right]}{d\left[\log(t)\right]} \tag{7}$$

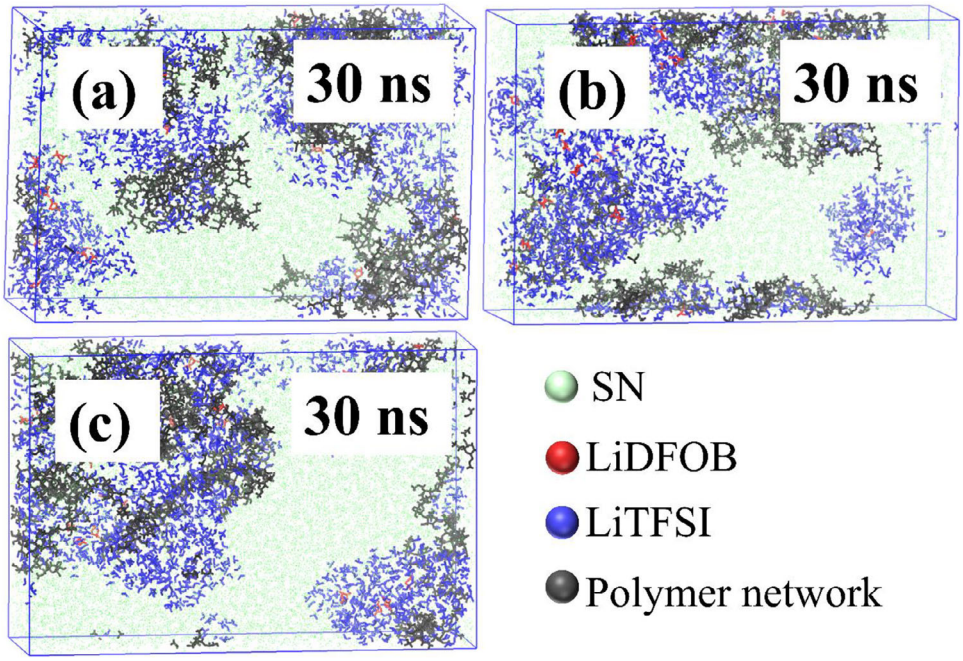


FIGURE 2 | Final electrolyte structure (a) CEC, (b) EC, (c) MC.

$$MSD = 6D_A t \tag{8}$$

Here, D_A represents the diffusion coefficient. $r_i(0)$ is the position at which the center of the atom at time 0. $r_i(t)$ is the position at which the center of the atom at time t.

2.3.4 | Ionic Conductivity Formulation

The ionic conductivity is a measure of a material's ability to conduct electric current through the movement of ions. In electrolytes, it represents how easily ions can move through the medium in response to an electric field. The Green-Kubo relationship is used to calculate ionic conductivity in MD simulations. The mathematical formulae of Green-Kubo are found in the previous literature [35, 56].

3 | Result and Discussion

3.1 | Nonbonded Interactions Through H-bonding

The average number of H-bonds between the polymer frame work to the other component of the eutectogel is shown in Figure 3. The average count of H-bonds between polymer and LiTFSI is presented in Figure 3a. Both CEC and MC have a comparable number of H-bonds during the dynamics. The cyano group is an electronegative group that is the CEC polymer matrix. However, the polar cyano functional group is responsible for introducing H-bonds capabilities mediated by its electronegative nitrogen moiety. MC has a relatively small methyl group with low steric hindrance. This, in turn, allows LiTFSI molecules to more easily access the remaining hydroxyl groups. However, EC has an ethyl group and thus has increased stearic hindrance in the cellulose backbone. The H-bonds between the polymer to the LiDFOB of the system are shown in Figure 3b. The show that forming more H-bonds with the LiDFOB. It forms more H-bonds with CEC because of the associated cyanoethyl groups. But the average number of H-bonds compared to Figure 3a is lower. The reduced presence of LiDFOB in the system explains this observation. Figure 3c illustrates the average H-bonds between the polymer network and SN, showing similar bond counts across all systems, indicating the eutectogel's stability. Figure 3d highlights intra- and interchain H-bonds within the polymer framework, revealing that the CEC framework forms more bonds due to cyanoethyl groups, while EC and MC exhibit comparable bonding. This indicates that the polymer network retains structural integrity, with these H-bonds serving as physical cross-links that enhance the gel network's stability.

3.2 | Structural Framework Through RDF and CN

Figure 4 shows the RDF between the lithium-ion present in LiTFSI to the different moieties of LiDFOB. The solvation shell value and *CNs* are given in Table S2 of the supplementary material. The electrostatic interactions that occur between moieties create the solvation shell. The highest RDF value, lowest solvation shell, and high *CN* are observed in Figure 4c, which suggests stronger interactions. This is due to the electronegative nature of fluorine atoms. However, the lowest solvation shell is observed in Figure 4b. The oxygen moiety of LiDFOB in the MC system shows this solvation shell, which is due to the simpler structure and smaller side groups present in the MC. Looking into the RDF results, the lithium-ion of LITFSI strongly interacts with LIDFOB moieties in the MC system. This is due to the MC's simpler structure and smaller side groups (-CH_3), creating a more organized polymer network with better-defined channels or pathways. The well-organized structure allows the lithium salts (both LiTFSI and LiDFOB) to distribute themselves more uniformly and come into closer proximity with each other. In contrast, the bulkier side groups in CEC (-CH_2CH_2CN) and

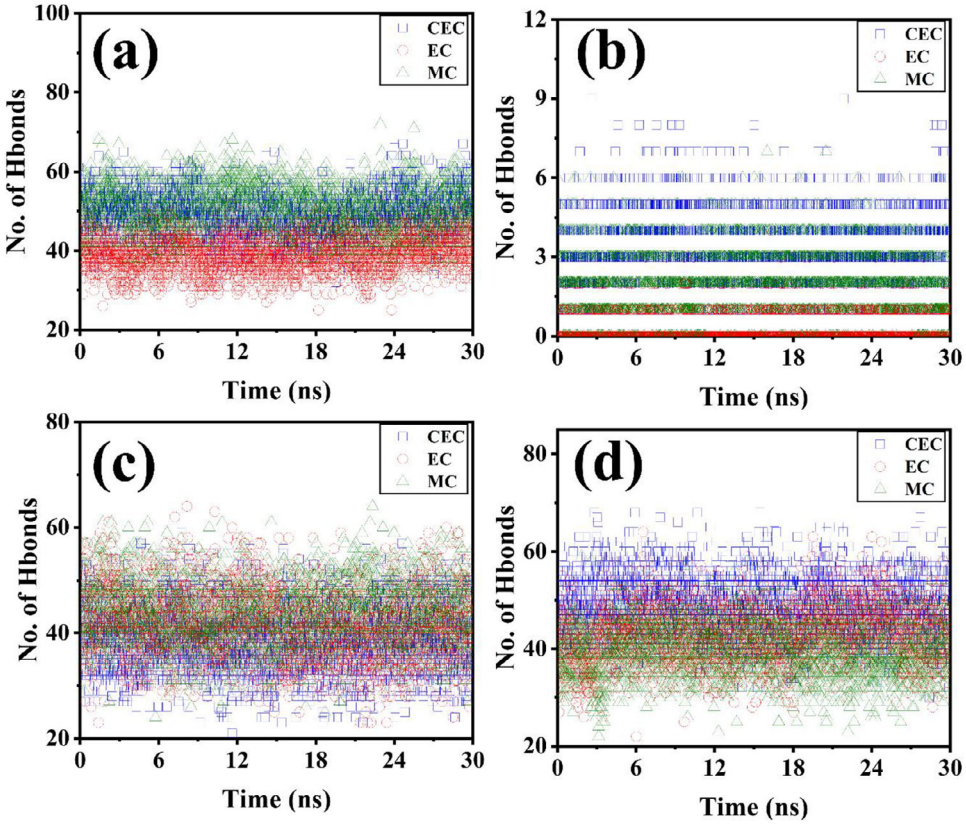


FIGURE 3 | Number of H-bonds (a) Polymer matrix to LiTFSI (b) Polymer matrix to LiDFOB (c) Polymer matrix to SN (d) Polymer matrix to the Polymer matrix.

EC (-CH_2CH_3) likely create a more disordered polymer matrix that disrupts the organization of lithium salts, leading to more scattered distributions and weaker coordination between LiTFSI and LiDFOB species. Figure S4 shows the RDF between the lithium-ion present in the LiTFSI to the different moiety of LiTFSI. The electrostatic interactions that occur between moieties create the solvation shell. In Figure S4b, the EC system shows higher *CN*. This signifies that lithium-ion is more surrounded by the oxygen moiety present in the LiTFSI compared to the other system. In Figure S4c, two consecutive solvation shells are observed due to the fluorine moiety present on both sides of the LiTFSI structure. However, one peak of RDF is higher, which suggests that the lithium-ion placed slightly shifted to one site. This figure shows the lowest solvation cell compared to the other figures. The fluorine moiety is the most electronegative atom, which suggests stronger interactions with lithium ions. Looking into Figure S4, the RDF values for the three systems are nearly equal, which suggests that the presence of the polymer matrix doesn't impact the interaction between the lithium-ion of LiTFSI to the other moiety of LiTFSI. Figure S5 of the supplementary material shows the RDF of lithium-ion of LiTFSI to the different moiety of polymer and the nitrogen moiety of SN. The electrostatic interactions that occur between moieties create the solvation shell. It is observed that the ether moiety of the cellulose backbone shows stronger interactions with the lithium-ion of LiTFSI. Comparing Figure 4 and Figures S4, S5, the RDF result shows stronger interactions for the lithium-ion present in the LiTFSI to another moiety of LiTFSI. This is due to the initial placement of lithium-ion in the LiTFSI structure. The higher RDF value is observed for the MC system, which is due to less steric hindrance.

Figure S6 shows the RDF between the lithium-ion of LiDFOB to the other moiety of the LiDFOB system. The solvation shell and *CN*s are given in Table S2 of the supplementary material. The highest RDF value is observed in Figure S6c. Due to the electronegativity of the fluorine moieties, the lithium ions strongly interact. It is observed that the EC and MC system shows similar RDF values, but the CEC system shows lesser RDF values. In the CEC system, the lithium ions are strongly interact with the carbonyl group present in the LiDFOB. Looking at Figure S4, the EC and MC systems show stronger interactions of lithium-ion except for the carbonyl group present in LiDFOB. The polarity of CEC arises from the electronegative cyano (-C≡N) groups, which create strong dipole moments. These groups promote dipole–dipole interactions with lithium ions, competing with the carbonyl group of LiDFOB for coordination. Additionally, the electron-withdrawing nature of cyano groups reduces the electron density on cellulose oxygen atoms, weakening lithium-ion interactions. CEC's hydrophilicity further reduces lithium-ion coordination by favoring solvation effects, unlike the less polar EC and MC. Figure 5 shows the RDF between the lithium-ion of LiDFOB to the other moiety of LiTFSI. The solvation shell and *CNs* are given in Table S2 of the supplementary material. The electrostatic interactions that occur between moieties create the solvation shell. The EC system shows a higher RDF compared to the other system. EC's moderate hydrophilicity balances lithium-ion solvation, promoting favorable diffusion and interactions with

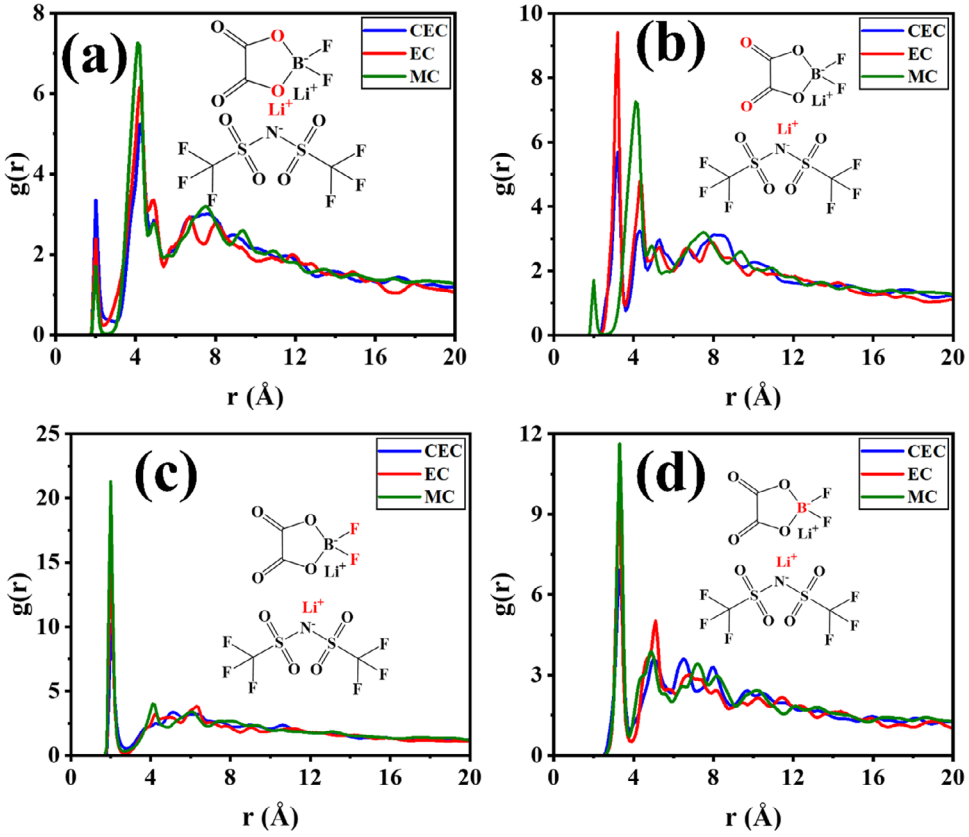


FIGURE 4 | RDF between the lithium-ion of LiTFSI to the other moiety of LiDFOB (a) ether moiety of LiDFOB (b) keto moiety of LiDFOB (c) fluorine moiety of LiDFOB (d) boron moiety of LiDFOB.

LiTFSI compared to the highly hydrophilic CEC. In contrast, MC, with less bulky substituents than EC, exhibits reduced polarity, which may not provide as favorable an environment for LiTFSI interactions. This combination of moderate hydrophilicity and polarity in EC results in higher RDF values, while the lower polarity of MC leads to comparatively weaker lithium-ion interactions with LiTFSI. Figure S7 of the supplementary material shows the RDF between the lithium-ion of LiDFOB to the different polymer moiety and SN. It is observed that the MC system is showing a higher RDF value, which is due to less stearic hindrance between the moieties. Comparing Figure 5 and Figures S6, S7, the highest RDF value and lowest solvation shell are observed in Figure 6. It is due to the placement of the lithium-ion in the LiDFOB system.

Figures S8 and S9 show the RDF between lithium ions and the oxygen moieties in LiTFSI at different time scales and system sizes. The production run extends from 30 to 300 ns. During this period, the solvation shell size and RDF values remain similar, as shown in Figure S8 of the SM. Figure S9 also shows that the solvation shell and RDF values stay the same when the system size is increased to four times the original. This indicates that changes in simulation time or the number of atoms do not affect the structural properties of the eutectogel system.

3.3 | Lithium-ion Dynamics Through MSD

Figure 6 shows the MSD of lithium-ion in the LiDFOB system. The higher MSD is observed in the EC system. The EC system's logarithmic slope value is more than unity, indicating the superdiffusive behavior. This behavior is observed due to the interactions of lithium ions and EC, which is shown in the RDF analysis. The polymer matrix provides a channel for faster ion transfer in the electrolyte system. However, the lowest MSD is observed in the CEC system. This lower value of the MSD is observed due to the higher interactions with the polymer matrix and electrolytes. The CEC system has a cyanoethyl group, which provides higher interactions with lithium. This is shown in the RDF results. The subdiffusive behavior of lithium ions is suggested by the logarithmic MSD slope value, which is less than unity. This signifies that the lithium ions are trapped in the eutectogel interaction core. The second-highest MSD is observed in the MC system, which is provided in Figure 6c. Initially, the MSD exhibits linear behavior until 11.8 ns, with a logarithmic MSD slope value of 0.97. This indicates subdiffusive behavior. This is caused by molecules being caught in the molecule's center. After this time span, the logarithmic slope of MSD declines till the dynamics end. This is due to the improved interactions between the polymer moiety and lithium, as seen in the RDF data.

Figure 7 shows the MSD of lithium-ion in the LiTFSI. The higher MSD is observed in the EC system. The logarithmic slope value of the EC system is unity, which signifies the diffusive behavior of lithium ions. The self-diffusion coefficient value is 1.44×10^{-12} $m^2.s^{-1}$. The lithium-ions are surrounded and stabilized by SN, TFSI ions, LiDFOB, and functional groups of EC. These interactions form a solvation shell around the ions, creating a localized environment that restricts rapid ion hopping.

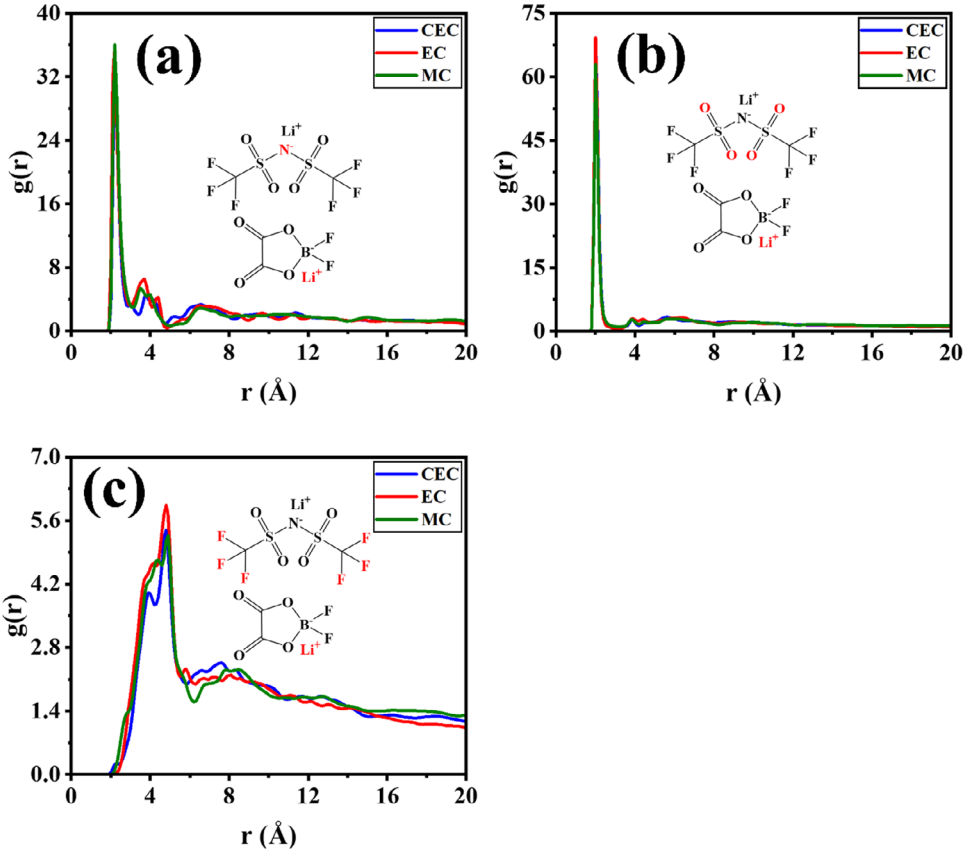


FIGURE 5 | RDF of the lithium-ion of LiDFOB to the other moiety of the LiTFSI (a) nitrogen moiety of LiTFSI, (b) oxygen moiety of LiTFSI (c) flurine moiety of LiTFSI.

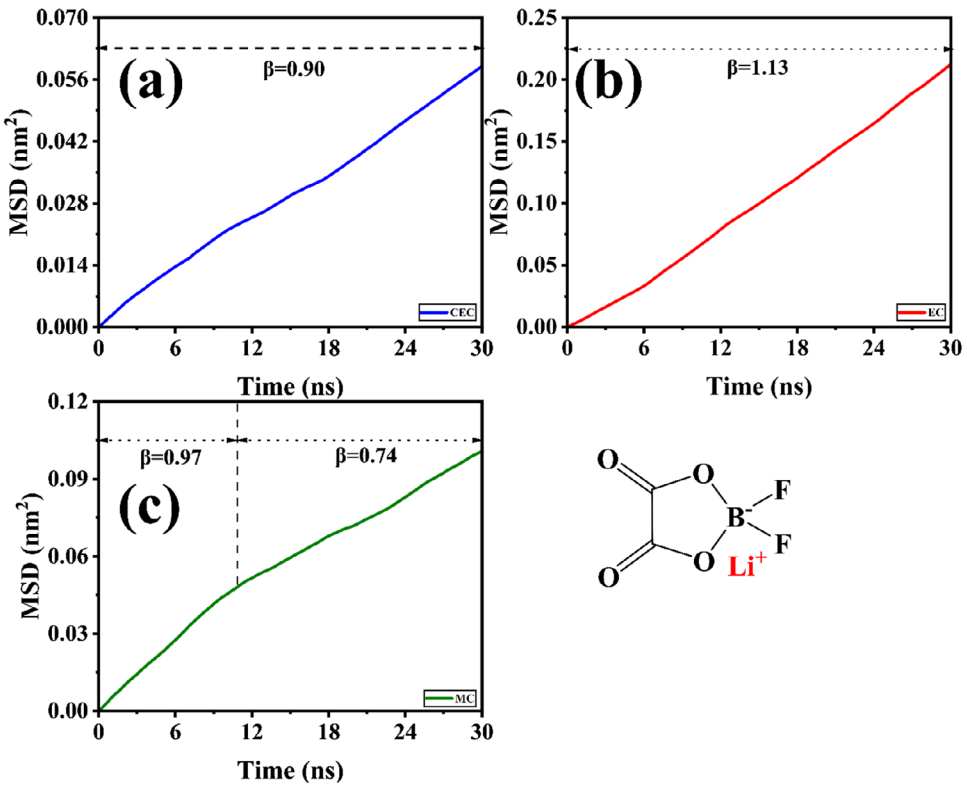


FIGURE 6 | MSD of lithium-ion of the LiDFOB (a) CEC (b) EC (c) MC.

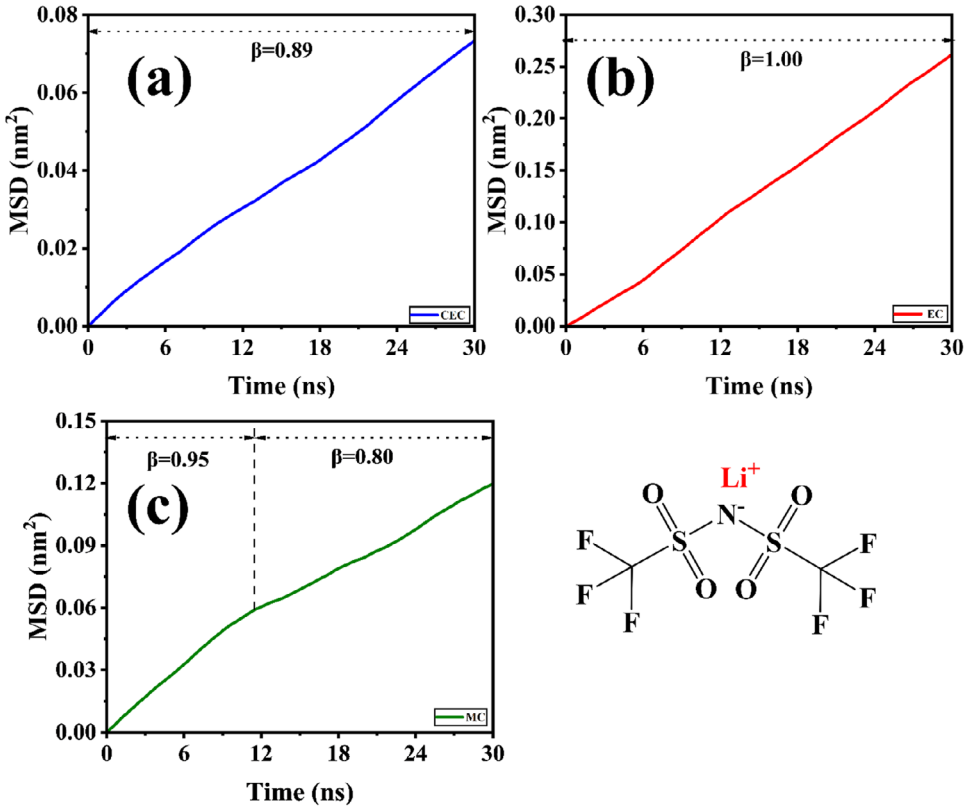


FIGURE 7 | MSD of lithium-ion of the LiTFSI (a) CEC (b) EC (c) MC.

The hydroxyl, ether moiety in EC and nitrile groups in SN have a strong affinity for lithium-ion, binding them within the matrix. As a result, the lithium ions diffuse uniformly, following a steady, self-diffusive pattern. The second-highest MSD was observed in the MC system. The MSD curve is divided into two regions based on their logarithmic MSD slope value. Initially, the MC system exhibits a linear MSD with a logarithmic slope value of 0.95 until 11.8 ns. The RDF results demonstrate that the lithium ions are trapped in the core of the cations and other moieties of the MC system throughout this duration. Following that, the MSD falls, and the dynamics become linear toward the end of the simulation time. The logarithmic slope value for this time period is 0.80, indicating that the moieties behaved subdiffusively. During this time period, the enhanced interactions with the MC matrix and other moieties of the electrolytes reduce the movement of lithium ions. The lowest MSD of lithium ions in LiTFSI is observed in the CEC system. The logarithmic MSD value for slope is less than unity, indicating subdiffusive behavior. Because of the improved interactions between the polymer moieties and DES with electrolytes, this subdiffusive behavior is observed. Comparing Figures 6 and 7, the highest MSD of lithium-ion is observed in LiTFSI compared to the lithium-ion of LiDFOB. This is due to a higher number of lithium ions present in the LiTFSI, which gives more accurate statistical averaging data.

Figures S10–S14 of SM shows the MSD of different moieties present in DES and the polymer matrix. The logarithmic slopes of MSD results suggest that the moieties are mainly present in subdiffusive and superdiffusive regions. The subdiffusive regions occur due to interactions between other moieties. As a result, the molecules are stuck on the solvation core. When molecules try to break the solvation core, the molecules show superdiffusive behavior. The superdiffusive happens owing to the molecular hopping process, which has been demonstrated in prior research [57].

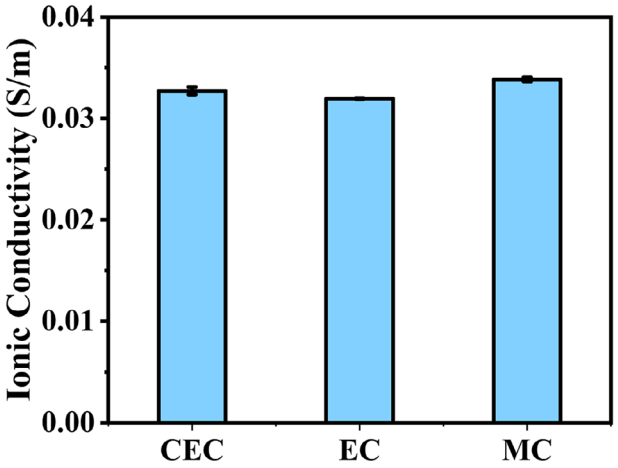


FIGURE 8 | Ionic conductivity of the different electrolytes.

3.4 | Ionic Nature Estimation

The ionic conductivity analysis of the three eutectogel systems reveals that the EC-based system exhibits the highest ionic conductivity of 3.39 S/cm, followed by MC at 3.27 S/cm, and CEC at 3.15 S/cm, as shown in Figure 8. The higher ionic conductivity in the EC-based system can be attributed to the optimal balance

between polymer chain flexibility and salt coordination. The -CH_2CH_3 side groups in EC provide sufficient spacing between polymer chains without introducing excessive steric hindrance, creating favorable pathways for ion transport, which is also seen in the MSD results. While the MC system's smaller -CH_3 moieties allow for tighter polymer packing, this might slightly restrict ion mobility compared to EC. The CEC system shows the lowest conductivity despite having polar cyanoethyl (-CH_2CH_2CN) moieties, likely because these bulkier side groups create more tortuous paths for ion migration and potentially stronger ion-polymer interactions that could impede ion transport. Based on the ionic conductivity results, the EC-based eutectogel system emerges as the most promising candidate for practical applications, as it provides the most efficient ion transport network. This enhanced conductivity suggests that the EC matrix creates an optimal balance between structural stability and ion mobility, making it the most suitable choice for applications requiring high ionic conductivity, such as in energy storage devices.

4 | Conclusion

All-atom MD simulation is employed to investigate the properties of DES-based solid-state electrolytes. The CEC system exhibits a higher number of H-bonds, indicating a more ordered structure compared to other systems. To analyze the lithium-ion interactions with its surroundings, RDF calculations are performed. The RDF results reveal that the EC and MC systems show stronger interactions between lithium ions and the polymer matrix, as well as the electrolyte environment. To further examine lithium-ion transport properties, MSD analysis is conducted. The MSD results indicate that lithium ions associated with LiTFSI exhibit higher mobility, reflected in greater MSD values, compared to lithium ions from LiDFOB. This difference arises due to the higher concentration of LiTFSI in the system, providing more statistically accurate results. Additionally, the lithium ions in the EC system display clear diffusive behavior, with a calculated self-diffusion coefficient of 1.44×10^{-12} $m^2.s^{-1}$. The ionic conductivity result suggests that the EC system has higher ionic conductivity, which is also validated by the MSD results. This suggests that the EC system facilitates better lithium-ion transport compared to other systems. Overall, the study highlights the influence of structural order, ion-polymer interactions, and ion transport properties on the performance of DES-based electrolytes. These findings provide critical insights into optimizing the design of cellulose-based electrolyte systems for advanced energy storage applications.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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